

Toxicogenomics II

Differential gene expression & pathway enrichment analysis

Dr. Marie Brasseur, AG Ecotoxicology

Differential expression analysis in R: DESeq2

1. Normalization
2. Dispersion estimation
3. GLM fit for each gene + testing

Differential expression analysis in R: DESeq2

1. Normalization -> for sequencing **depth** & library **composition**
2. Dispersion estimation
3. GLM fit for each gene + testing

	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Sample 6
Gene 1	1500	1650	1350	0	10	47
Gene 2	4504	5896	2587	10256	11589	9584
Gene 3	3587	5895	2598	657	894	585
Gene 4	2569	3512	2014	4512	3520	2597
Sum	12160	16953	8549	15425	16013	12813

Highly expressed genes consume reads

Low coverage

Differential expression analysis in R: DESeq2

1. Normalization
2. Dispersion estimation -> gene expression heterogeneity across samples
3. GLM fit for each gene + testing

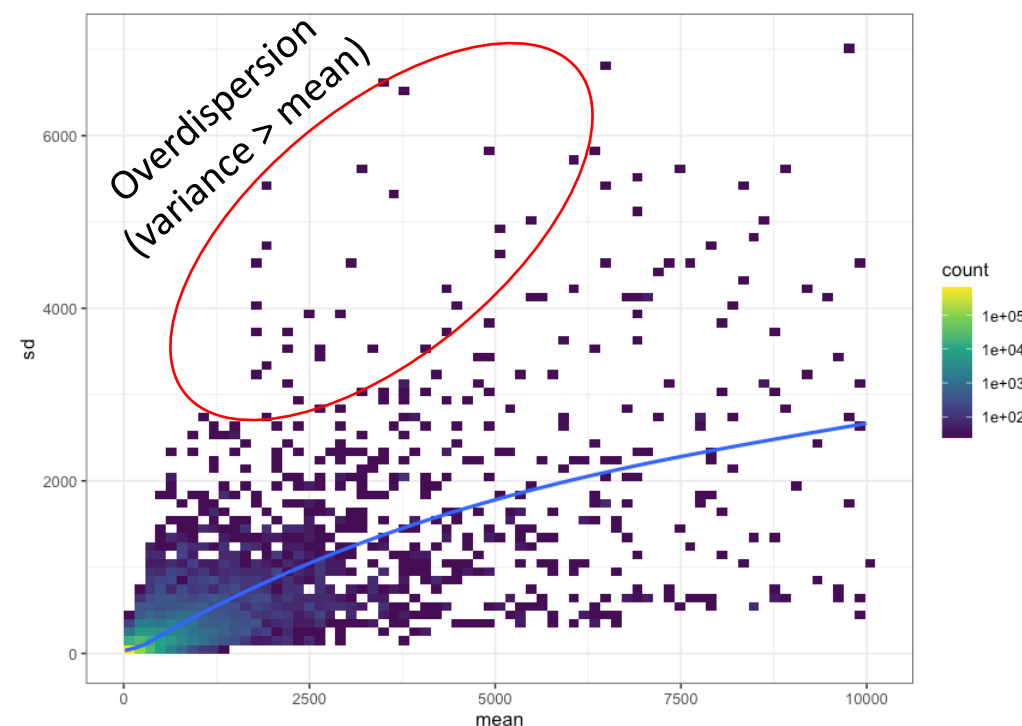
Parameter of the negative binomial distribution

Differential expression analysis in R: DESeq2

1. Normalization
2. Dispersion estimation -> gene expression heterogeneity across samples
3. GLM fit for each gene + testing

Mean-variance relationship for gene counts

	sample_1	sample_2	sample_3	sample_4	sample_5	sample_6		mean	variance
gene_1	8287	7027	4202	1535	12867	11808		7621	4349
gene_2	114	806	221	56	235	409		307	273
gene_3	967	926	315	18	1807	1746		963	727
gene_4	6	189	72	16	44	70		66	66
gene_5	2153	1916	2349	878	2171	1767		1872	529
gene_6	433	2296	603	209	978	1430		992	771

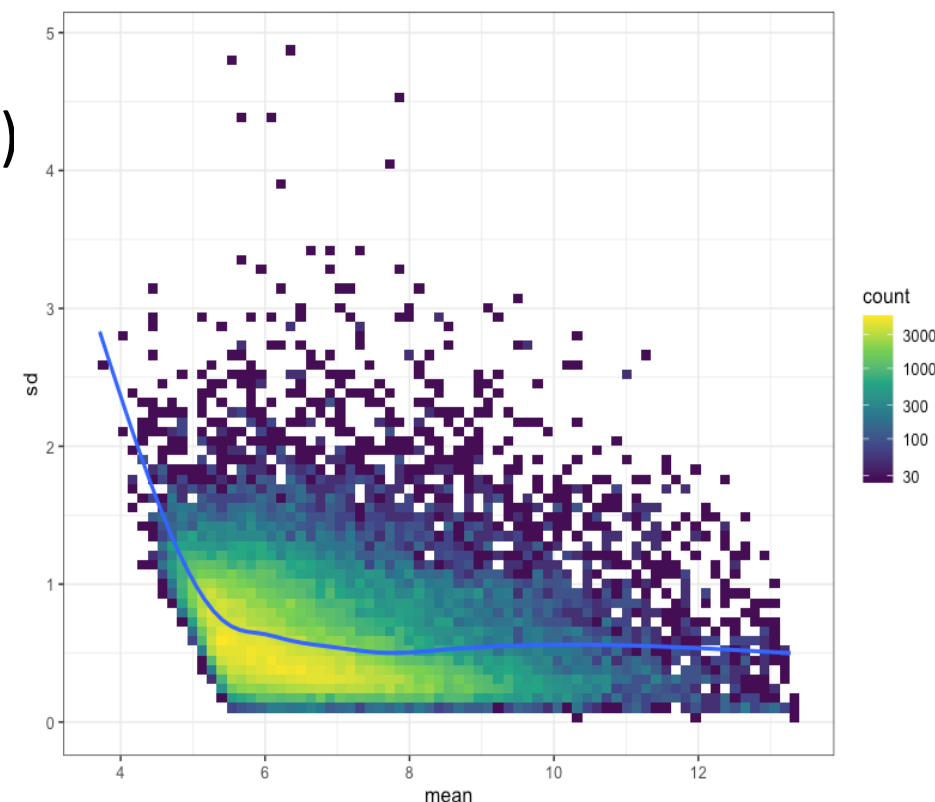


Differential expression analysis in R: DESeq2

1. Normalization
2. Dispersion estimation -> gene expression heterogeneity across samples
3. GLM fit for each gene + testing

Mean-variance relationship for **log₂(gene counts)**

	sample_1	sample_2	sample_3	sample_4	sample_5	sample_6	mean	variance
gene_1	13	13	12	11	14	14	13	1
gene_2	7	10	8	6	8	9	8	1
gene_3	10	10	8	4	11	11	9	3
gene_4	3	8	6	4	5	6	5	2
gene_5	11	11	11	10	11	11	11	1
gene_6	9	11	9	8	10	10	10	1



Differential expression analysis in R: DESeq2

1. Normalization
2. Dispersion estimation -> gene expression heterogeneity across samples
3. GLM fit for each gene + testing

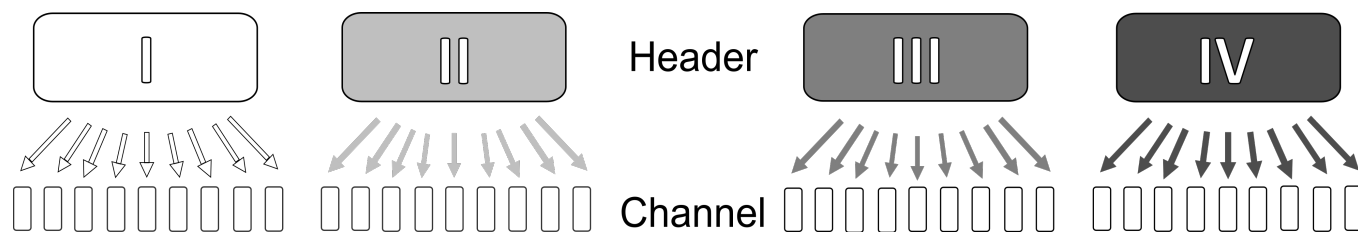
Parameter of the negative binomial distribution to fit overdispersed data

$$\text{Variance} = \text{mean} + \text{dispersion} * \text{mean}^2$$

Dispersion: inversely related to mean but directly related to variance

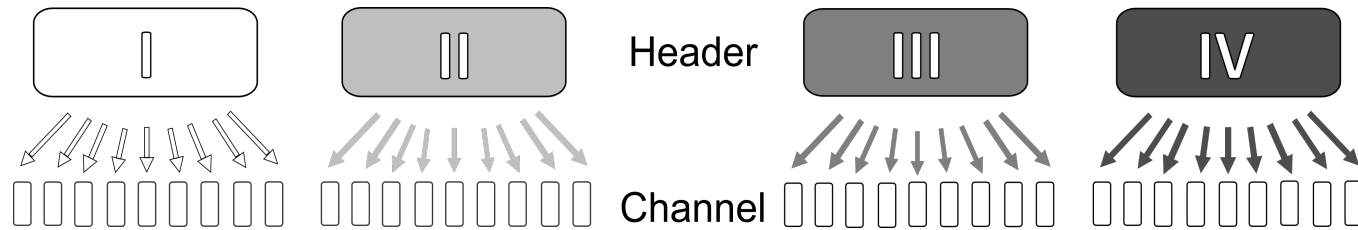
The data set

- Experimental design



The data set

- Experimental design



- Low insecticide (0.2 $\mu\text{g/L}$)
- Med. insecticide (2 $\mu\text{g/L}$)
- High insecticide (20 $\mu\text{g/L}$)

→ Three non-target taxa

Lepidostoma basale



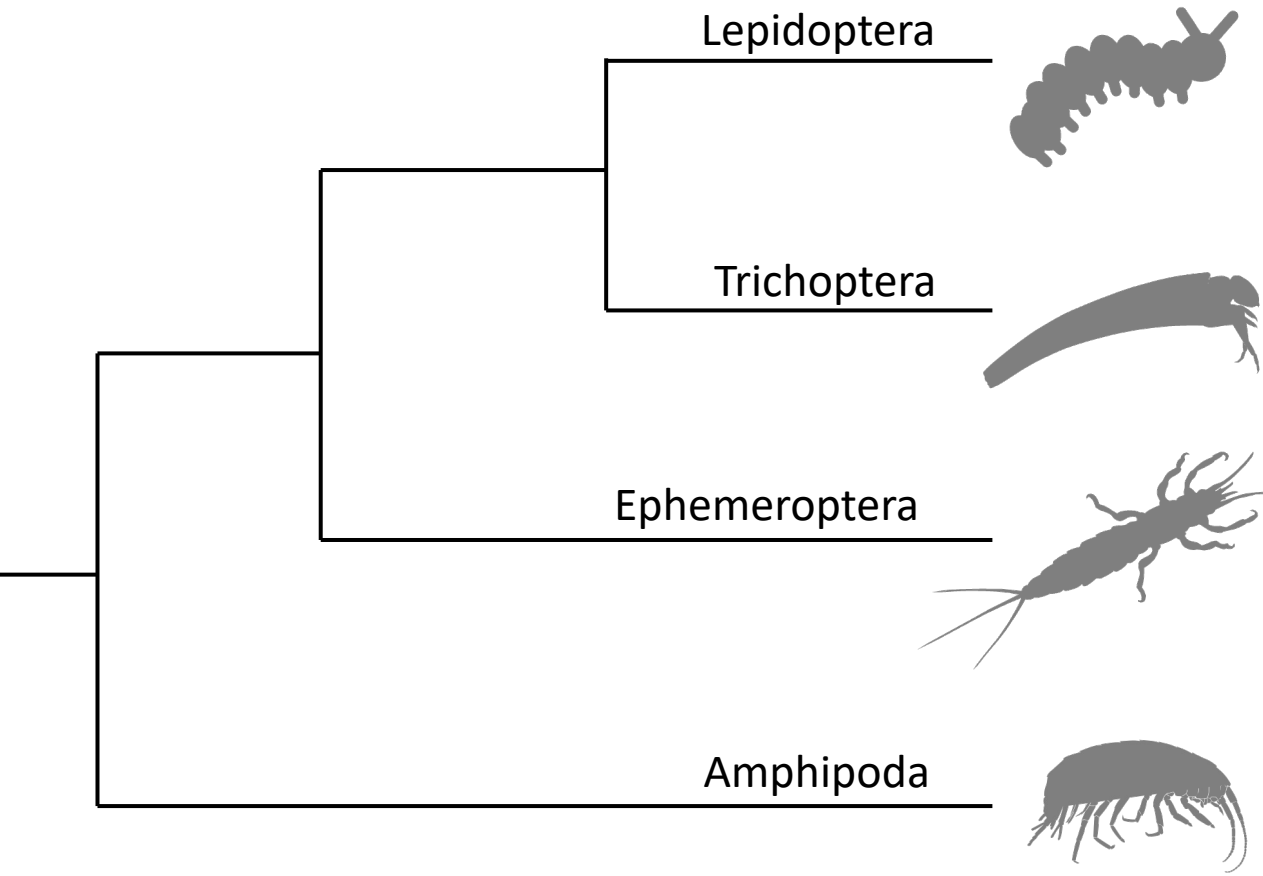
Ephemera danica



Gammarus pulex



Phylogenetic & ecological spectrum



Lepidostoma basale



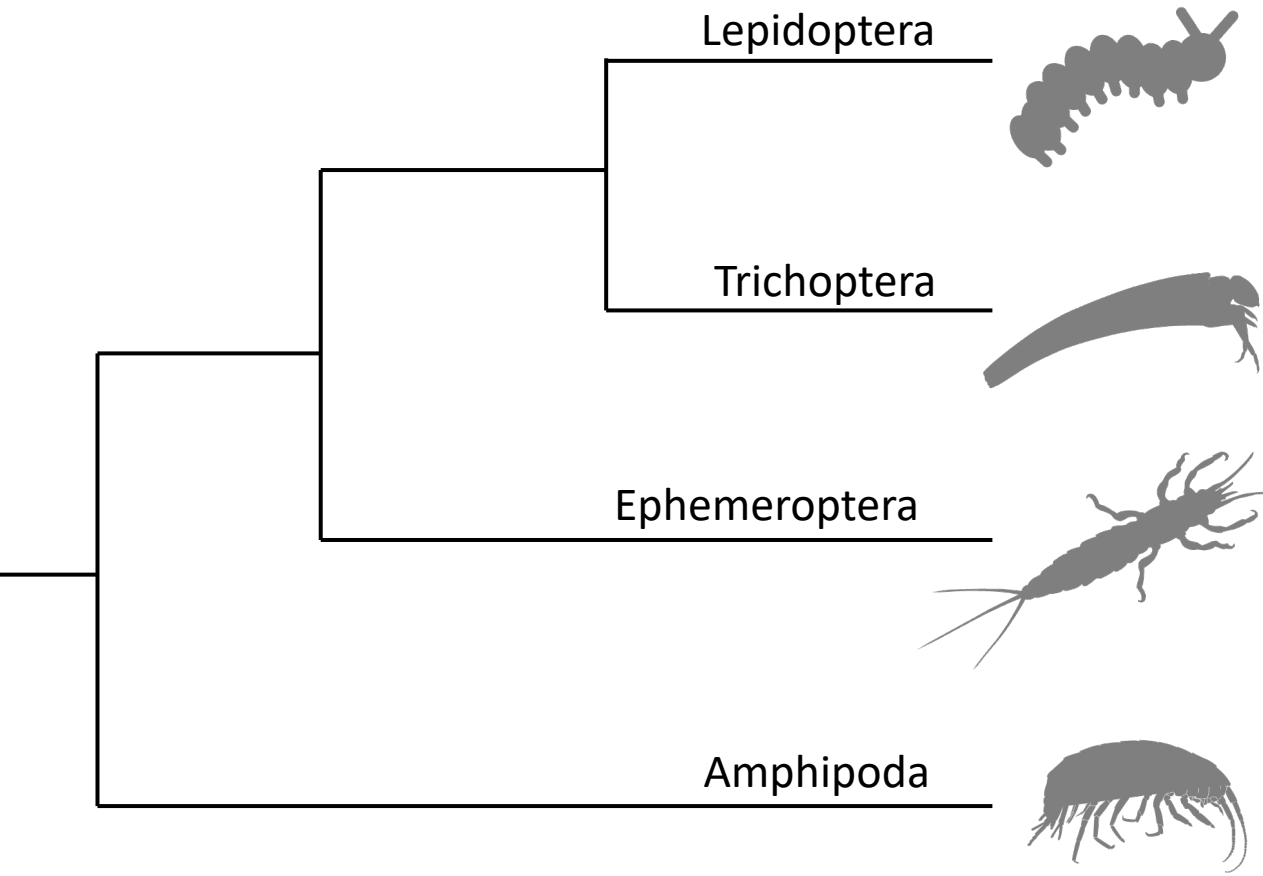
Ephemera danica



Gammarus pulex



Phylogenetic & ecological spectrum



Lepidostoma basale



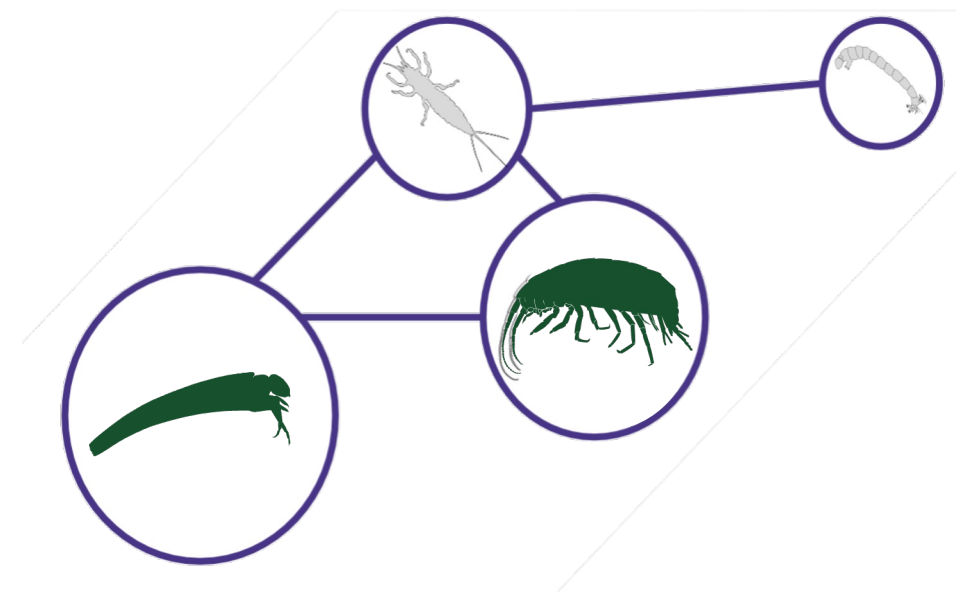
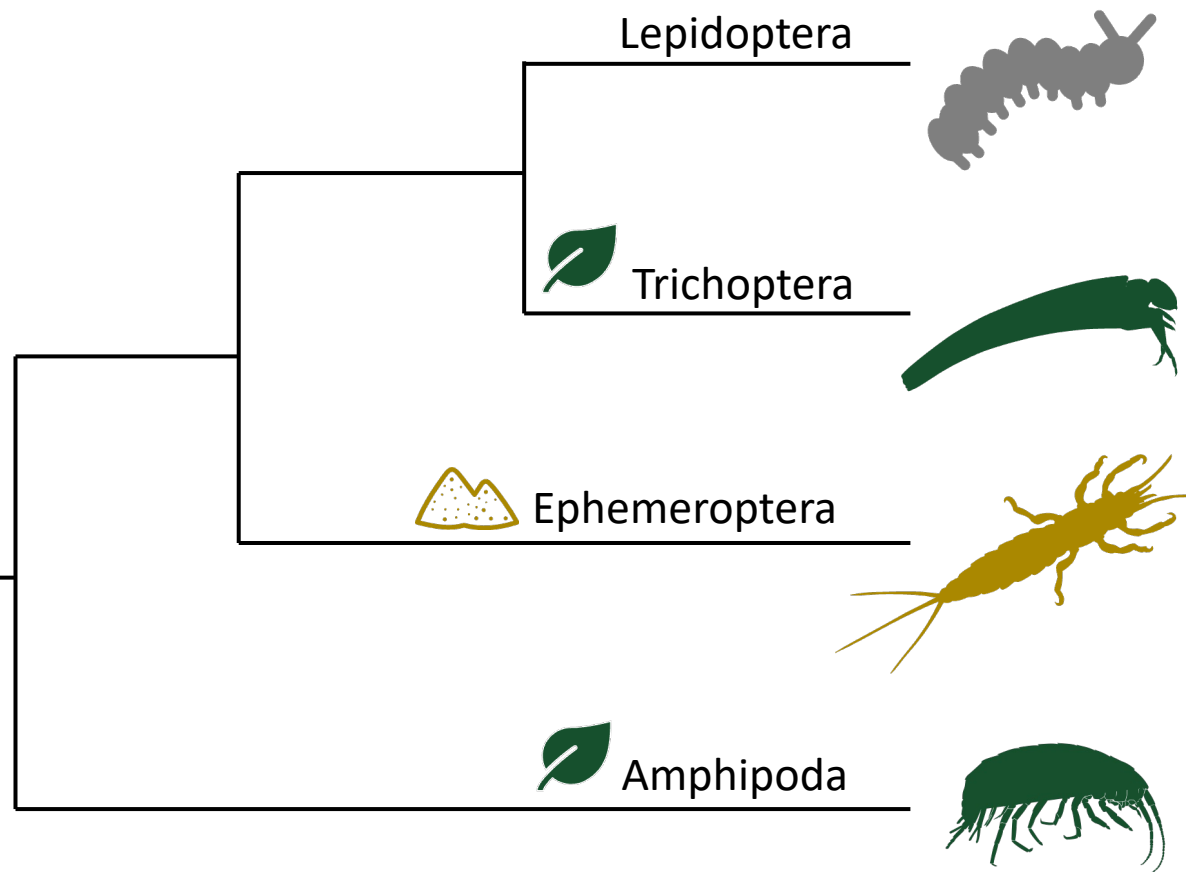
Ephemera danica



Gammarus pulex



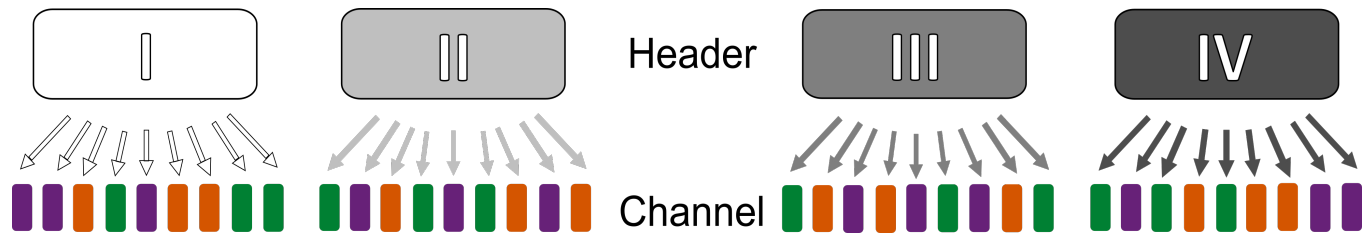
Phylogenetic & ecological spectrum



Does interspecific competition modulate stress responses?

The data set

- Experimental design



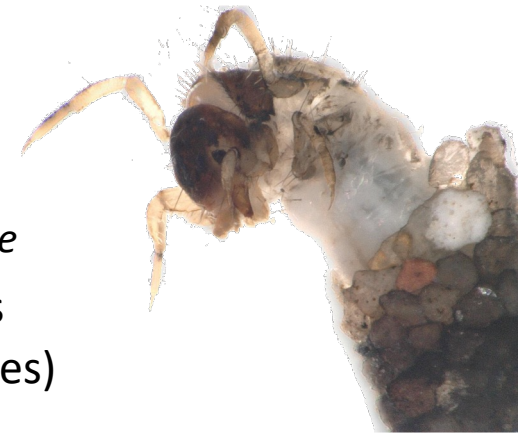
- Low insecticide (0.2 µg/L)
- Med. insecticide (2 µg/L)
- High insecticide (20 µg/L)

- *E. danica* + *L. basale*
- *E. danica* + *G. pulex*
- *E. danica* + *L. basale* + *G. pulex*

Full factorial design (3 replicates)

48 hours of exposure

Lepidostoma basale
(234 specimens
24 RNA-seq libraries)



Ephemera danica

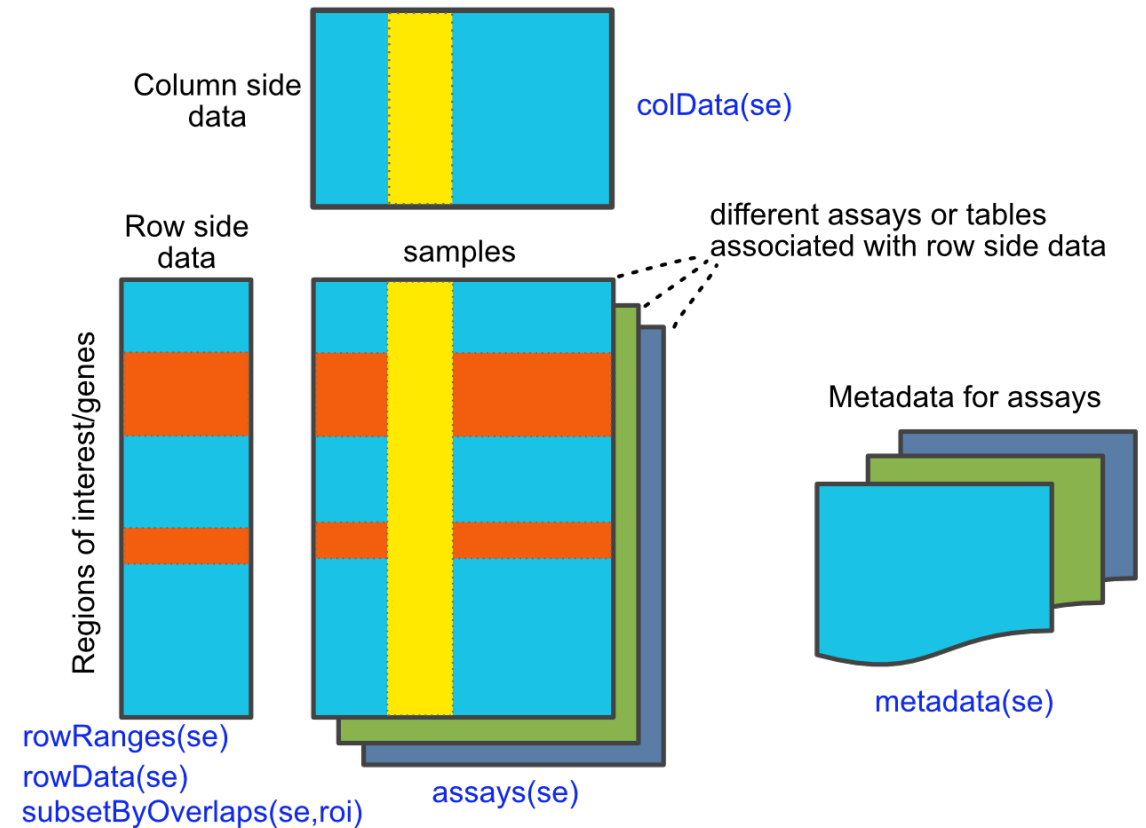


Gammarus pulex



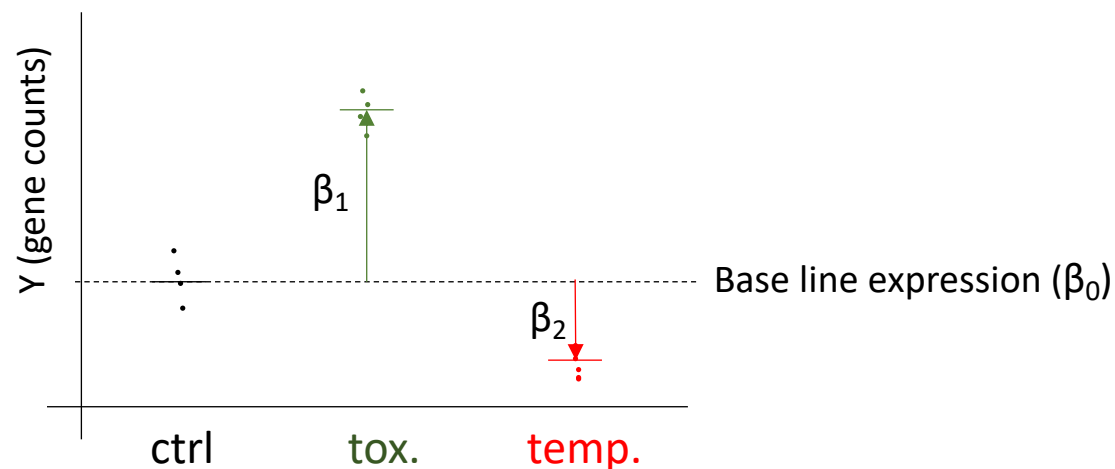
Coding session

1. Data manipulation
2. DESeq2
 - 3.1 Normalization of counts
 - 3.2 Dispersion estimation
 - 3.3 GLM fit
4. Hypothesis testing
5. Visualisation, PCA
6. Optional: pathway analysis



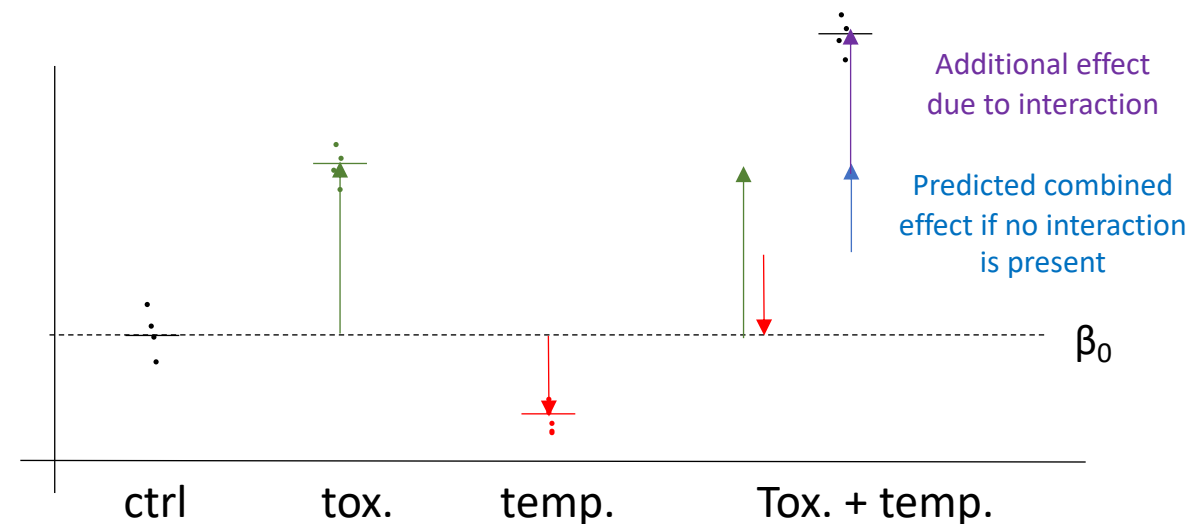
Interpretation of predictors to get the desired contrasts

$$Y_i = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n \varepsilon$$



$Y \sim \text{tox.} + \text{temp.}$
(predictors both categorical)

- No interaction term specified = independent effects

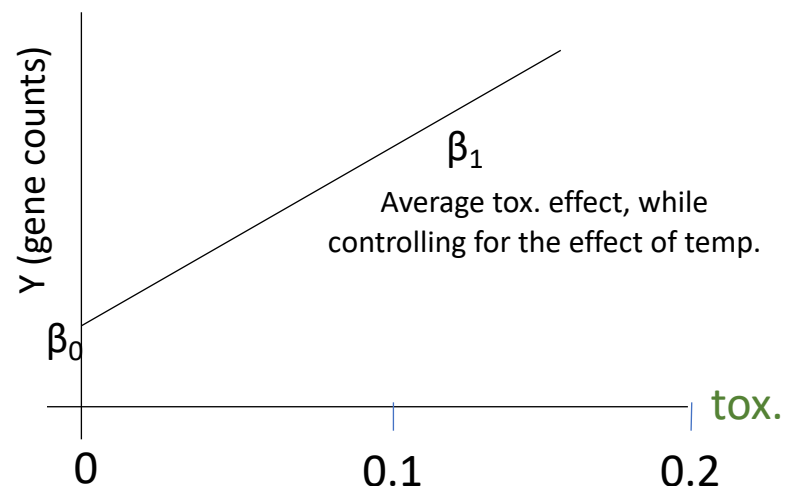


$Y \sim \text{tox.} * \text{temp.}$
(predictors both categorical)

- Interaction = effect of tox. depends of on the level of temp (&vice versa)

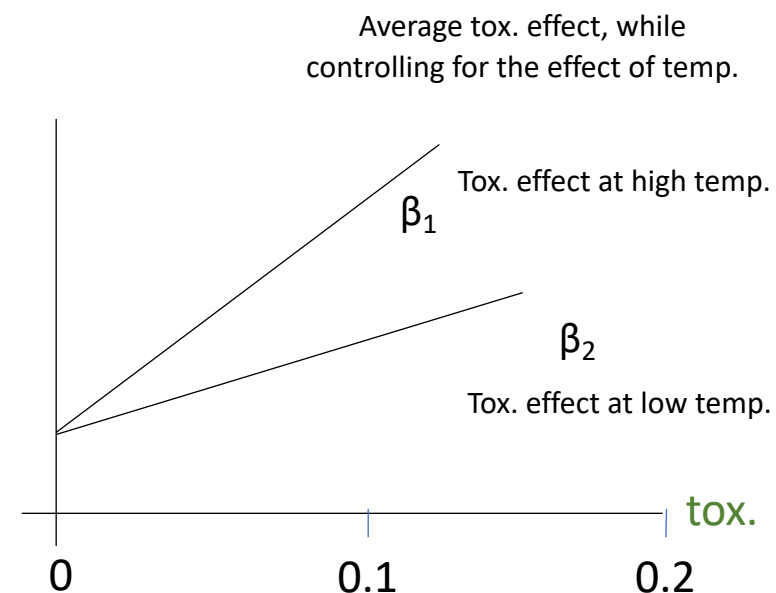
Interpretation of predictors to get the desired contrasts

$$Y_i = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n \varepsilon$$



$Y \sim \text{tox.} + \text{temp.}$
(tox. predictor numerical)

- No interaction term specified = independent effects



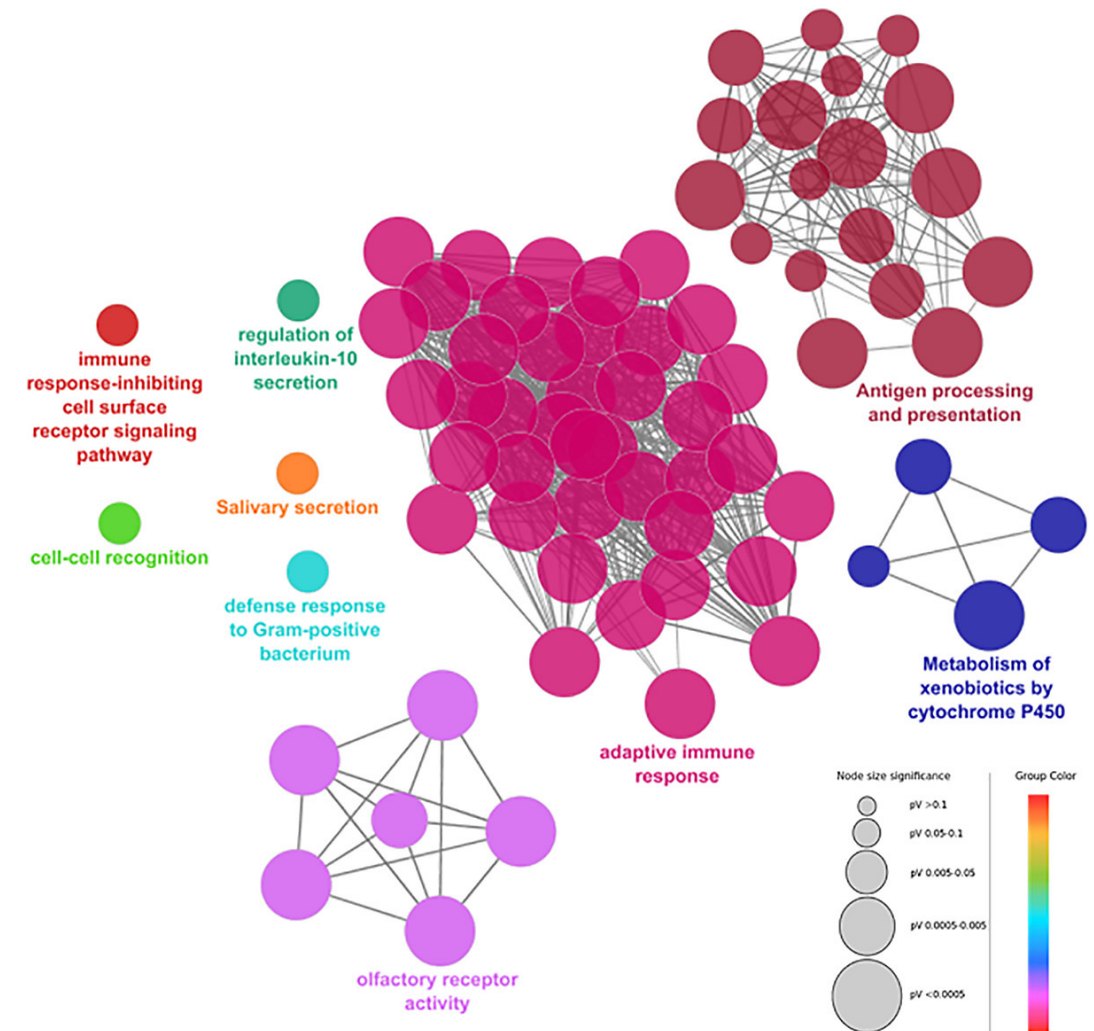
$Y \sim \text{tox.} * \text{temp.}$
(tox. predictor numerical)

- Interaction = effect of tox. depends of on the level of temp (&vice versa)

Pathway analysis

- Finally, we want to know which molecular pathways are **stimulated** and **suppressed**

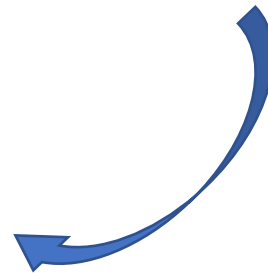
-> Functional profiling of **upregulated** and **downregulated** genes



Functional annotation

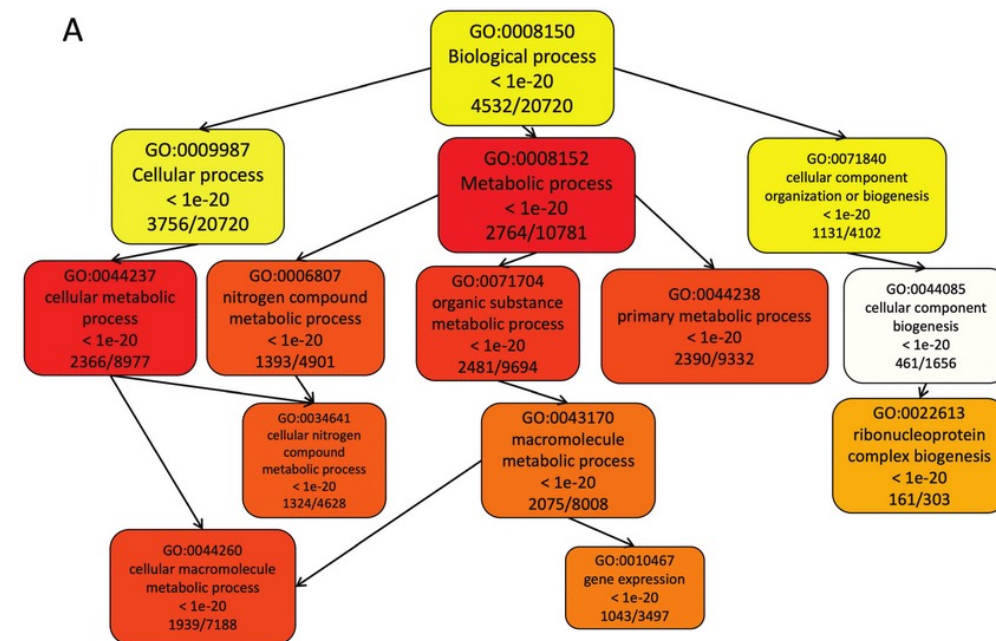
- Reference genomes/transcriptomes can be annotated
- > Compare sequences in references with sequences of genes with known function

In databases such as UniProt



Functional annotation

- Function can be encoded as e.g. GO terms
-> describe what genes do – in a standardized, consistent way across species
- GO terms classify gene functions into three main categories: **B**iological **P**rocess, **M**olecular **F**unction, **C**ellular **C**omponent
- The terms are arranged in a **hierarchical structure**: Broad terms at the top (e.g., "metabolism"), specific terms underneath (e.g., "lipid metabolism")



Functional enrichment

- Which functions are overrepresented in your gene set (e.g. toxicant-induced genes)
- > e.g., Fisher's Exact Test

	Differential Expression	NO Differential Expression	Total
IN Transcription Elongation	12	3	15
NOT IN Transcription Elongation	3	12	15
Total	15	15	30

